AMS 526: Numerical Linear Algebra for Computational and Data Sciences

Lecture 12: Givens Rotation; Least Squares Problems; Conditioning of Least Squares Problems

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Outline

1. **Givens Rotations**
2. Linear Least Squares Problems (NLA §11)
3. Conditioning of Least Squares Problems (NLA §18)

Givens Rotations

- Instead of using reflection, we can rotate $\boldsymbol{x}$ to obtain $\|\boldsymbol{x}\|\boldsymbol{e}_1$.
- A Givens rotation

$$\boldsymbol{R} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

  rotates $\boldsymbol{x} \in \mathbb{R}^2$ counterclockwise by θ.
- Choose θ to be the angle between $(x_i, x_j)^\mathsf{T}$ and $(1, 0)^\mathsf{T}$, and we have

$$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x_i \\ x_j \end{bmatrix} = \begin{bmatrix} \sqrt{x_i^2 + x_j^2} \\ 0 \end{bmatrix},$$

  where

$$\cos\theta = \frac{x_i}{\sqrt{x_i^2 + x_j^2}}, \quad \sin\theta = \frac{-x_j}{\sqrt{x_i^2 + x_j^2}}.$$

Givens QR

- Introduce zeros in a column bottom-up, one zero at a time, such as

$$\begin{bmatrix} \times & \times & \times \\ \times & \times & \times \\ \times & \times & \times \\ \times & \times & \times \\ \times & \times & \times \end{bmatrix} \xrightarrow{(4,5)} \begin{bmatrix} \times & \times & \times \\ \times & \times & \times \\ \times & \times & \times \\ \mathbf{x} & \mathbf{x} & \mathbf{x} \\ \mathbf{0} & \mathbf{x} & \mathbf{x} \end{bmatrix} \xrightarrow{(3,4)} \begin{bmatrix} \times & \times & \times \\ \times & \times & \times \\ \mathbf{x} & \mathbf{x} & \mathbf{x} \\ \mathbf{0} & \mathbf{x} & \mathbf{x} \\ \mathbf{0} & \times & \times \end{bmatrix} \cdots$$

- To zero a_{ij}, left-multiply the matrix F with $F_{i:i+1,i:i+1}$ being a rotation matrix and $F_{kk} = 1$ for $k \neq i, i+1$.
- The flop count of Givens QR is $3mn^2 - n^3$, which is about 50% more expensive than Householder triangularization.

Adding a Row

- Suppose $A \in \mathbb{R}^{m \times n}$ with $m \geq n$, and A has full rank.
- Let $\tilde{A} = \begin{bmatrix} A_1 \\ z^\top \\ A_2 \end{bmatrix}$, where $A = \begin{bmatrix} A_1 \\ A_2 \end{bmatrix}$ and $z^\top$ is a new row inserted.
- Obtain $\tilde{A} = \tilde{Q}\tilde{R}$ from $A = QR$ efficiently using Givens rotations.
  - Suppose $A = \begin{bmatrix} A_1 \\ A_2 \end{bmatrix} = \begin{bmatrix} Q_1 \\ Q_2 \end{bmatrix} R$.
  - Then $\tilde{A} = \begin{bmatrix} A_1 \\ z^\top \\ A_2 \end{bmatrix} = \begin{bmatrix} 0 & Q_1 \\ 1 & 0^\top \\ 0 & Q_2 \end{bmatrix} \begin{bmatrix} z^\top \\ R \end{bmatrix}$.
  - Perform a series of Givens rotations $\tilde{R} = U_n^\top \dots U_2^\top U_1^\top \begin{bmatrix} z^\top \\ R \end{bmatrix}$, and then

    $$\tilde{Q} = \begin{bmatrix} 0 & Q_1 \\ 1 & 0^\top \\ 0 & Q_2 \end{bmatrix} U_1 U_2 \dots U_n.$$
  - Updating $\tilde{R}$ costs $3n^2$ flops, and updating $\tilde{Q}$ costs $6mn$ flops.

Adding a Column

- Suppose $A \in \mathbb{R}^{m \times n}$ with $m \geq n$, and A has full rank.
- Let $\tilde{A} = \begin{bmatrix} A_1 & z & A_2 \end{bmatrix}$, where $A = \begin{bmatrix} A_1 & A_2 \end{bmatrix}$ and z is a new column.
- Obtain $\tilde{A} = \tilde{Q}\tilde{R}$ from $A = QR$ efficiently using Givens rotations.
  - Suppose $A = \begin{bmatrix} A_1 & A_2 \end{bmatrix} = Q \begin{bmatrix} R_1 & R_2 \end{bmatrix}$.
  - Then $\tilde{A} = \begin{bmatrix} A_1 & z & A_2 \end{bmatrix} = Q \begin{bmatrix} R_1 & w & R_2 \end{bmatrix}$, where $w = Q^\top z$.
  - Perform a series of Givens rotations $\tilde{R} = U_{k+1} \cdots U_n \begin{bmatrix} R_1 & w & R_2 \end{bmatrix}$, where U_n performs on rows n and $n-1$, U_{n-1} performs on rows $n-1$ and $n-2$, etc.
  - Thus, $\tilde{Q} = QU_n^\top \cdots U_{k+1}^\top$.
  - It takes $\mathcal{O}(mn)$ time overall.

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2. **Linear Least Squares Problems (NLA §11)**
3. Conditioning of Least Squares Problems (NLA §18)

Linear Least Squares Problems

- An overdetermined system of equations $Ax \approx b$, where A has more rows than columns and has full rank, in general has no solutions.
- An example application is polynomial least squares fitting.
- In general, we minimize the residual $r = b - Ax$.
- In terms of the 2-norm, we obtain the linear least squares problem where given $A \in \mathbb{R}^{m \times n}$, $m \geq n$, and $b \in \mathbb{R}^m$, we find $x \in \mathbb{R}^n$ such that $\|b - Ax\|_2$ is minimized.
- If A has full rank, the minimizer x is the solution to the normal equation

$$A^\top A x = A^\top b$$

or in terms of the *pseudoinverse* A^+,

$$x = A^+ b, \quad \text{where } A^+ = (A^\top A)^{-1} A^\top \in \mathbb{R}^{n \times m}.$$

Geometric Interpretation

- Ax is in $\mathrm{range}(A)$, and the point in $\mathrm{range}(A)$ closest to b is its orthogonal projection onto $\mathrm{range}(A)$.
- The residual r is then orthogonal to $\mathrm{range}(A)$, and hence $A^\top r = A^\top(b - Ax) = 0$.
- Ax is the orthogonal projection of b, where $x = A^+ b$, so $P = AA^+ = A(A^\top A)^{-1}A^\top$ is an orthogonal projection.

Solution of Least Squares Problems

- One could solve $A^\top A x = A^\top b$ directly using Cholesky factorization.
  - This is unstable, but it is efficient if $m \gg n$ ($mn^2 + \frac{1}{3}n^3$ flops).
- A more robust approach is to use $A = \hat{Q}\hat{R}$.
  - We can project b onto $\mathrm{range}(A)$ by $P = \hat{Q}\hat{Q}^\top$, and therefore $\hat{Q}\hat{R}x = \hat{Q}\hat{Q}^\top b$.
  - Left-multiply by $\hat{Q}^\top$ and we get $\hat{R}x = \hat{Q}^\top b$.

Least squares via QR factorization

Compute reduced QR factorization $A = \hat{Q}\hat{R}$
Compute vector $c = \hat{Q}^\top b$
Solve upper-triangular system $\hat{R}x = c$ for x

- Computation is dominated by the QR factorization ($2mn^2 - \frac{2}{3}n^3$ flops).
- Question: If Householder QR is used, how do we compute $\hat{Q}^\top b$?

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- Answer: Compute $Q^\top b$ (where Q is from the full QR factorization) and then take the first n entries of $Q^\top b$.

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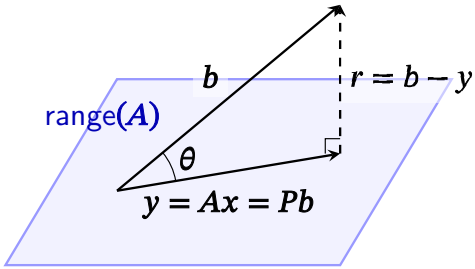
Four Conditioning Problems

- Least squares problem: Given $A \in \mathbb{R}^{m\times n}$ with full rank and $b \in \mathbb{R}^m$, find

$$\min_{x\in\mathbb{R}^n} \|b - Ax\|.$$

- Its solution is $x = A^+ b$. Another quantity is $y = Ax = Pb$, where $P = AA^+$.



- Consider A and b as input data, and x and y as output. We then have four conditioning problems.

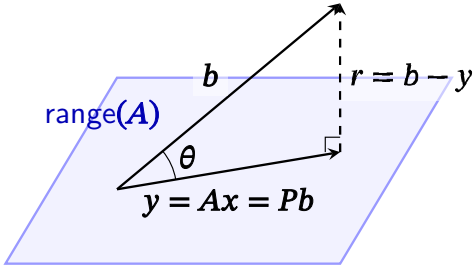
Input \ Output	y	x
b	$\kappa_{b\to y}$	$\kappa_{b\to x}$
A	$\kappa_{A\to y}$	$\kappa_{A\to x}$

- These conditioning problems are important and subtle.

Some Prerequisites

- We focus on the second column, namely $\kappa_{b\to x}$ and $\kappa_{A\to x}$.
- However, understanding $\kappa_{b\to y}$ and $\kappa_{A\to y}$ is a prerequisite.

- Three quantities (all in 2-norms):
  - Condition number of A:
    $\kappa(A) = \|A\|\|A^+\| = \sigma_1/\sigma_n$.
  - Angle between b and y: $\theta = \arccos \frac{\|y\|}{\|b\|}$
    $(0 \le \theta \le \pi/2)$.
  - Orientation of y with range(A): $\eta = \frac{\|A\|\|x\|}{\|y\|}$
    $(1 \le \eta \le \kappa(A))$.



Sensitivity of y to Perturbations in b

- Intuition: The larger θ is, the more sensitive y is in terms of relative error.
- Analysis: $y = Pb$, so

$$\kappa_{b\to y} = \frac{\|P\|}{\|y\|/\|b\|} = \frac{\|b\|}{\|y\|} = \frac{1}{\cos\theta},$$

  where $\|P\| = 1$.

Input \ Output	y	x
b	$\frac{1}{\cos\theta}$	
A		

- Question: For which nonzero perturbation δb, with $\|\delta b\|$ fixed, is the supremum in the definition of $\kappa_{b\to y}$ attained?

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- Question: For which nonzero perturbation δb, with $\|\delta b\|$ fixed, is the supremum in the definition of $\kappa_{b\to y}$ attained?
- Answer: When δb is in $\mathrm{range}(A)$.

Sensitivity of x to Perturbations in b

- Intuition: It depends on how sensitive y is to b, and how y lies within range(A).
- Analysis: $x = A^+ b$, so

$$\kappa_{b\to x} = \frac{\|A^+\|}{\|x\|/\|b\|} = \|A^+\| \frac{\|b\|}{\|y\|} \frac{\|y\|}{\|x\|} = \|A^+\| \frac{1}{\cos\theta} \frac{\|A\|}{\eta} = \frac{\kappa(A)}{\eta \cos\theta},$$

where $\eta = \|A\|\|x\|/\|y\|$.

Input \ Output	y	x
b	$\frac{1}{\cos\theta}$	$\frac{\kappa(A)}{\eta\cos\theta}$
A		

Sensitivity of x to Perturbations in b

- Assume $\cos\theta = \mathcal{O}(1)$, $\kappa_{b\to x} = \dfrac{\kappa(A)}{\eta\cos\theta}$ can lie anywhere between 1 and $\mathcal{O}(\kappa(A))$!
- Question: For which nonzero perturbation δb, with $\|\delta b\|$ fixed, is the supremum in the definition of $\kappa_{b\to x}$ attained?

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- Question: For which nonzero perturbation δb, with $\|\delta b\|$ fixed, is the supremum in the definition of $\kappa_{b\to x}$ attained?
- Answer: When δb is in the subspace spanned by left singular vectors corresponding to the smallest singular values.
- Question: What if A is a nonsingular matrix?

Sensitivity of x to Perturbations in b

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- Question: For which nonzero perturbation δb, with $\|\delta b\|$ fixed, is the supremum in the definition of $\kappa_{b\to x}$ attained?
- Answer: When δb is in the subspace spanned by left singular vectors corresponding to the smallest singular values.
- Question: What if A is a nonsingular matrix?
- Answer: $\kappa_{b\to x}$ can lie anywhere between 1 and $\kappa(A)$!

Sensitivity of y and x to Perturbations in A

- The relationships are nonlinear, because $\text{range}(A)$ changes due to δA.
- Intuitions:
  - The larger θ is, the more sensitive y is in terms of relative error.
  - Tilting of $\text{range}(A)$ depends on $\kappa(A)$.
  - For x, it depends on where y lies within $\text{range}(A)$.

Input \ Output	y	x
b	$\frac{1}{\cos\theta}$	$\frac{\kappa(A)}{\eta\cos\theta}$
A	$\leq \frac{\kappa(A)}{\cos\theta}$	$\leq \kappa(A) + \frac{\kappa(A)^2\tan\theta}{\eta}$

- For the second row, bounds are not necessarily tight.
- Assume $\cos\theta = \mathcal{O}(1)$, then $\kappa_{A\to x}$ can lie anywhere between $\kappa(A)$ and $\mathcal{O}\left(\kappa(A)^2\right)$.

Condition Numbers of Linear Systems

- The linear system $Ax = b$ for a nonsingular $A \in \mathbb{R}^{m \times m}$ is a special case of least squares problems, where $y = b$.
- If $m = n$, then $\theta = 0$, so $\cos\theta = 1$ and $\tan\theta = 0$.

Input \ Output	y	x
b	1	$\kappa(A)/\eta$
A	—	$\leq \kappa(A)$